

Comparison of Percolation Test Results and Estimated Hydraulic Conductivities for Mollisols and Alfisols

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ABSTRACT

Percolation tests were performed on an array of biosequences, topo-drainage sequences, and stage of profile development sequences of soils in Iowa and results were compared to estimated subsoil hydraulic conductivities from Soil Interpretation Records. Within a biosequence, Mollisols had higher percolation rates and lower coefficients of variation than Alfisols. The difference in permeability was attributed to higher subsoil clay content, higher subsoil bulk density, more continuous argillans, and stronger subsoil consistence in Alfisols. High coefficients of variation for percolation rates in Alfisols were attributed to large diameter root channels, lateral flow of water on sloping sites, and increased distinctness of subsoil horizon boundaries. Percolation rates of Mollisols commonly exceed Soil Interpretation Records' estimates of hydraulic conductivity by a factor of two or three while percolation rates of Alfisols approximate estimated subsoil hydraulic conductivity. In soils without argillic horizons, percolation test results of topo-drainage sequences are similar and reflect similar morphological characteristics. Soil Interpretation Records assign slower estimated hydraulic conductivity to some topo-drainage sequences as drainage becomes more restrictive. Percolation rates decreased in topo-drainage sequences of soils with argillic horizons as drainage became more restrictive, even though morphological characteristics were quite similar. Differences in percolation rates within topo-drainage sequences may reflect errors inherent in the procedure. Additional permeability data is needed to quantify and more accurately estimate subsoil hydraulic conductivity. Criteria used to establish estimated hydraulic conductivity in Soil Interpretation Records should be applied consistently among biosequences and topo-drainage sequences.

DESPITE ITS LIMITATIONS (Franzmeier et al., 1964; McGauhey and Krone, 1967; Bouma, 1971, 1973; Winneberger, 1972; Mellen, 1973; Healy and Laak, 1973) the percolation test remains the most widely used method of predicting site suitability and determining absorption field size for on-site sewage disposal systems (OSDS) (Winneberger, 1984). More precise and accurate methods are available to determine soil permeability but limitations of technical expertise, time, and expense often restrict their usage (Anderson and Bouma, 1977; Fritton et al., 1986).

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Numerous studies have indicated a relationship between percolation rates and soil characteristics (McGauhey and Krone, 1967; Derr et al. 1969; Persinger and Yahner, 1970; Healy and Laak, 1973; Lewis, 1975; Barbarick et al., 1976).

The percolation test has been described as a three-dimensional infiltration measurement, generally performed under unsaturated soil conditions with a variable hydraulic head (Chan, 1976; Hill, 1966). Bouma (1971) indicated that percolation rates varied considerably from saturated hydraulic conductivities because the percolation of water through the bottom and sidewalls of test holes is determined by flow patterns with undefined boundary conditions in a relatively large volume of soil with a hydraulic head greater than unity.

Percolation rates have been compared to hydraulic conductivity values determined by the double-tube method, (Bouma, 1971); core method, (Winneberger, 1974); and shallow well pump-in method (Fritton et al., 1986). Linear regression analysis of the resulting pairs of data have yielded prediction equations to estimate hydraulic conductivity from percolation rate. A site evaluation model developed by Fritton et al. (1982; 1986) utilizes percolation rates to calculate saturated hydraulic conductivity values.

Morris et al. (1962), Olson (1964), Huddleston and Olson (1967), and Bouma (1974) have suggested that soil survey information can be more accurate than percolation test results in assessing suitability of soils for OSDS. Permeability of the soil is a prerequisite for designing and sizing the absorption field after site suitability is established, however. Soil survey reports provide estimates of saturated hydraulic conductivity and assign permeability classes for selected horizons or depths for each soil series (Table 1). Saturated hydraulic conductivity is estimated on the basis of soil morphology and physical properties (Soil Conservation Service, 1978). Examination of estimated hydraulic conductivity values in soil survey reports in Iowa, leads one to conclude that the influence of native vegetation and natural soil drainage class on subsoil hydraulic conductivity, was either disregarded or not applied consistently. Percolation rates have been found to be correlated with native vegetation and natural soil drainage class in Iowa (Bicki, 1977; Luce, 1973).

The objectives of this study were to: (i) evaluate

Table 1. Permeability class and estimated hydraulic conductivity from SOI-5 Soil Interpretation Records (Soil Conservation Service, 1985).

Permeability class	Estimated hydraulic conductivity, $\text{m s}^{-1} \times 10^{-4}$
Ver slow	<0.42
Slow	0.42–1.41
Moderately slow	1.41–4.23
Moderate	4.23–14.1
Moderately rapid	14.1–42.3
Rapid	42.3–141.0
Very rapid	>141.0

percolation test results for an array of soils in Iowa representing biosequences, topo-drainage sequences, and stage of development sequences of soils, and (ii) to compare percolation rates to estimated subsoil hydraulic conductivities assigned to those soils in the SCS SOI-5 Soil Interpretation Records (Soil Conservation Service, 1985).

MATERIALS AND METHODS

Model of Study

Soils were selected on the basis of a model that provided a systematic framework for comparison of Iowa soils. The model was in part derived from previous studies of Ulrich (1949), Hutton (1950), White and Riecken (1955), Cain and Riecken (1958), Corliss (1958), and Richardson and Riecken (1977). It consists of (i) three classes of vegetation, (ii) four natural soil drainage classes, (iii) three parent material classes and (iv) four "stage of profile development classes."

The three vegetation classes, grouped by soil series comprise nine (i.e. five complete and four incomplete) biosequences found in Iowa (Table 2). Each biosequence has constraints of natural drainage, parent material, and stage of profile development. The four drainage classes, group by soil series comprise eight (i.e. seven complete and one incomplete) topo-drainage sequences (Table 3). Each topo-drainage sequence has constraints of native vegetation, parent material, and stage of profile development. Each soil series was assigned to one of four stages of profile development classes based on maximum clay content in the B horizon. This approach has historically been applied to loess-derived soils in Iowa (Richardson and Riecken, 1977). Soils in de-

velopmental class I have lower clay maximum in the B horizon than their vegetation and drainage class analogs in class II, II < III, and III < IV.

Percolation Test

The percolation test procedure was a modified falling-head auger hole method (Miller et al., 1981). Four to six holes 0.15 m in diam were bored to a depth of 0.9 m with a bucket auger or mechanically operated screw-type auger. Holes were spaced evenly within a 2- to 4-m² area (Fig. 1). The bottom and sides of the holes were roughed to remove smeared soil surfaces and loose soil material was removed. The sidewalls of each hole were lined with screening mesh (~1 mm openings) wire and approximately 0.05 m of noncalcareous pea gravel was added to the bottom of the holes. Tap water was added to test holes as necessary to maintain at least 0.18 m of water above the gravel for a period of not less than 8 h. The water level was then adjusted to a depth of 0.18 m above the gravel. Percolation rates were determined with an average water head of 0.15 m (Fig. 1). During the measurement period, the water level was allowed to decrease to a minimum depth of 0.13 m above the gravel before additional water was added to reestablish the water level to a depth of 0.18 m. The drop in water level was recorded for a 3- to 4-h period and the percolation rate for each test hole was calculated by averaging all readings taken during the final 3 h of testing or until a constant rate was obtained. Percolation test results for each soil series represents the mean of all test hole percolation rates at all sites. The number of test sites (*n*) for each soil series is given in Tables 4 and 6. Percolation rates were found to be log normally distributed. The base 10 logarithm of percolation rates was evaluated against the base 10 logarithm of estimated hydraulic conductivity by simple direct comparison. An analysis of variance and *F*-test (Snedecor and Cochran, 1967) was utilized to test significance of percolation test results within biosequences and topo-drainage sequences.

Hydraulic Conductivity

Estimated hydraulic conductivity of the soil at a depth of 0.76- to 1.0-m was obtained from Soil Conservation Service (1985) SOI-5 Soil Interpretation Records. The percolation test auger holes were bored to a depth of 0.9 m which is approximately the midpoint of the 0.76- to 1.0-m soil depth.

Table 2. Soil series arranged by biosequence, native vegetation, parent material, natural drainage class, and developmental stage.

Biosequence	Native vegetation			Parent material	Natural drainage class	Developmental stage†
	Prairie	Forest/Prairie	Forest			
Tama	Tama (Typic Argiudoll)	Downs (Mollic Hapludalf)	Fayette (Typic Hapludalf)	Loess	Well	II
Otley	Otley (Typic Argiudoll)	Ladoga (Mollic Hapludalf)	Clinton (Typic Hapludalf)	Loess	Moderately well	III
Grundy	Grundy (Aquic Argiudoll)	Pershing (Aquic Hapludalf)	Weller§ (Aquic Hapludalf)	Loess	Somewhat poor	IV
Mahaska	Mahaska var.‡ (Aquic Argiudoll)	Givin (Udolic Ochraqulf)		Loess	Somewhat poor	III
Taintor	Taintor (Typic Argiaquoll)	Rubio (Mollic Albaqualf)		Loess	Poor	III
Clarion	Clarion (Typic Hapludoll)	Lester (Mollic Hapludalf)	Hayden (Typic Hapludalf)	Glacial till	Well	I
Nicollet	Nicollet (Aquic Hapludoll)	LeSueur var.‡ (Udolic Ochraqulf)		Glacial till	Somewhat poor	II
Webster	Webster (Typic Haplaquoll)	Cordova (Typic Argiaquoll)	Ames (Typic Albaqualf)	Glacial till	Poor	II
Kenyon	Kenyon (Typic Hapludoll)	Bassett (Mollic Hapludalf)		Loamy material over till	Moderately well	I

† Developmental stages were, in part, derived from previous studies by Ulrich (1949), Hutton (1950), White and Riecken (1955), Cain and Riecken (1958), and Richardson and Riecken (1977).

‡ Clay content in subsoil is slightly below the range established in the official series description.

§ Weller series is moderately well-drained.

Table 3. Soil series arranged by topo-drainage sequence, natural drainage class, parent material, native vegetation, and developmental stage.

Topo-drainage sequence	Natural drainage class				Parent material	Native vegetation	Developmental stage†
	Well	Moderately well	Somewhat poor	Poor			
Galva	Galva (Typic Hapludoll)		Primghar (Aquic Hapludoll)	Marcus (Typic Haplaquoll)	Loess	Prairie	I
Sharpsburg		Sharpsburg (Typic Argiudoll)	Macksburg (Aquic Argiudoll)	Winterset (Typic Argiaquoll)	Loess	Prairie	III
Otley		Otley (Typic Argiudoll)	Mahaska var.‡ (Aquic Argiudoll)	Taintor (Typic Argiaquoll)	Loess	Prairie	III
Ladoga		Ladoga (Mollic Hapludalf)	Givin (Udolic Ochraqualf)	Rubio (Mollic Albaqualf)	Loess	Forest/Prairie	III
Clarion	Clarion§ (Typic Hapludoll)		Nicollet (Aquic Hapludoll)	Webster (Typic Haplaquoll)	Glacial till	Prairie	II
Lester	Lester§ (Mollic Hapludalf)		LeSueur var.‡ (Udolic Ochraqualf)	Cordova (Typic Argiaquoll)	Glacial till	Forest/Prairie	II
Hayden	Hayden§ (Typic Hapludalf)			Ames (Typic Albaqualf)	Glacial till	Forest	II
Kenyon		Kenyon (Typic Hapludoll)	Floyd (Aquic Hapludoll)	Clyde (Typic Haplaquoll)	Loamy material over till	Prairie	I

† Developmental stages were in part, derived from previous studies by Ulrich (1949), Hutton (1950), White and Riecken (1955), Cain and Riecken (1958), and Richardson and Riecken (1977).

‡ Clay content in subsoil is slightly below the range established in the official series description.

§ Developmental Stage I soil.

Particle Size, Bulk Density, and Morphological Features

Particle size analysis on the <2 mm soil fraction was determined by the pipette method (Kilmer and Alexander, 1949). Bulk density was determined by the tube method (Goddard et al., 1971) and core method (Blake, 1965). Morphologic characteristics, occurrence of argillans, soil structure, distinctness of horizon boundaries, and consistence were described using standard soil characterization methods and terminology (Soil Survey Staff, 1981). An analysis of variance and *F*-test (Snedecor and Cochran, 1967) was utilized to test significance of clay content and bulk density determinations where percolation tests were performed on more than one site.

RESULTS AND DISCUSSION

Biosequences

The prairie (Mollisol) end-members of the Tama, Otley, Mahaska, Clarion, and Nicollet biosequences, developmental stage I, II, and III soils, have significantly higher percolation rates and display a general trend of lower coefficients of variation than transition (Alfisol) members and forest (Alfisol) end-members of their respective biosequences (Table 4). A lack of significant difference in percolation rates between transition members and forest end-members of biosequences suggests that: (i) morphology of the B horizon in transition soils of these biosequences is more similar to forest end-member soils than prairie end-member soils or (ii) the high coefficients of variation in forest and transition members require large differences in order to be statistically different. Transition soils have been interpreted as having Late Holocene encroachment of forest vegetation on soils developed under prairie vegetation (Richardson and Riecken, 1977). Morphological differences that could account for the lower percolation rates of transition members and forest end-members of biosequences include: (i) a higher subsoil clay content; (ii) higher subsoil bulk density; (iii) thicker and more continuous argillans in the subsoil; and (iv) a stronger subsoil consistence (Table 5). Percolation test results have been correlated with subsoil clay content, bulk density, and subsoil consistence (Derr et al., 1969; Persinger and Yahner, 1970; Lewis, 1975; Barbarick et al., 1976).

Prairie (Mollisol) end-members of biosequences typically have the lowest variation in percolation test results (Table 4). The higher coefficients of variation in transition (Alfisol) members and forest (Alfisol) end-members of biosequences, are attributed to the presence of isolated large-diameter root channels in some

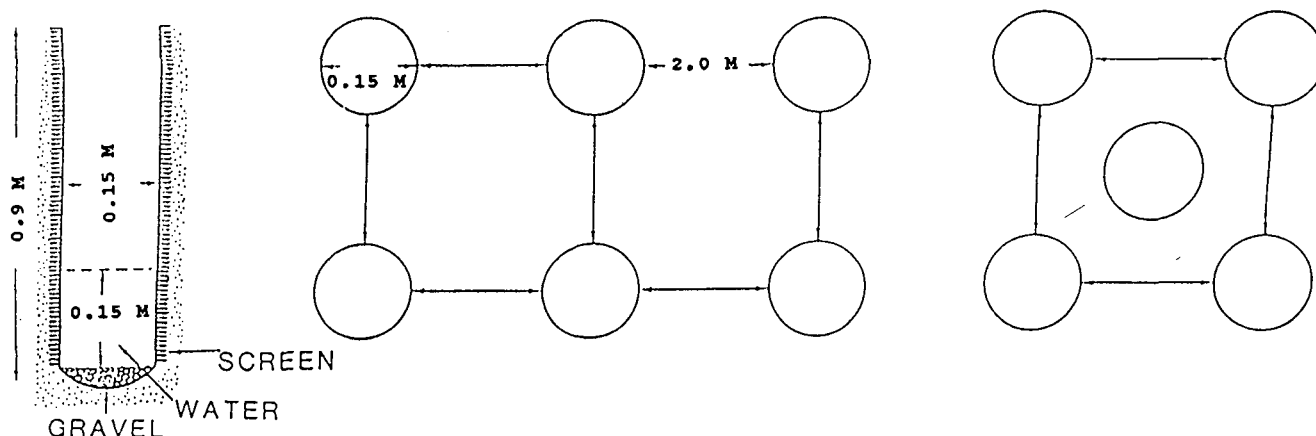


Fig. 1. Cross-sectional view of percolation test hole and arrangement of percolation test holes.

test holes. Test holes containing large-diameter root channels were found to transmit water at appreciably greater rates. Higher coefficients of variation in transition and forest end-members of biosequences located on sloping sites are related to lateral subsurface flow of water from upslope test holes to downslope test holes. The increased coefficients of variation in transition members and forest end-members of biosequences may also be related to the distinctness of subsoil horizon boundaries (Table 5). Alfisols were found to have more distinct horizon boundaries in the subsoil than Mollisols. Placement of the test zone in the same horizon becomes more critical as distinctness of boundaries increase because physical properties such as clay content, consistence, and bulk density change over a shorter vertical distance.

No significant differences were found in percolation rates among prairie (Mollisol) end-members and transition (Alfisol) members of the Taintor, Webster, and Kenyon biosequences. This suggests that B horizon morphology of transition members of these biosequences may be more similar to prairie end-members than forest end-members. Clay content, occurrence and thickness of argillans, and soil consistence are similar in the subsoil of prairie end-members and transition members of the Taintor, Webster, and Kenyon biosequences but additional sites should be examined to statistically evaluate morphological characteristics of these biosequences (Table 7). Coefficients of variation are lower in prairie end-members than transition members of the Taintor and Kenyon biosequences and are similar in the Webster biosequence, however.

The Grundy biosequence, a developmental stage IV

Table 4. Percolation test results and estimated hydraulic conductivity for selected biosequences and stage of profile development sequences.

Soil series†	n	Percolation test results			Estimated hydraulic conductivity‡
		Mean	Standard deviation	Coefficient of variation	
		$\text{m s}^{-1} \times 10^{-4}$		%	$\text{m s}^{-1} \times 10^{-4}$
Tama	2	31.10a*	4.72	15	4.23-14.10
Downs	2	9.17b	2.22	24	4.23-14.10
Fayette	3	9.17b	4.72	51	4.23-14.10
Otley	2	26.10a	9.44	36	4.23-14.10
Ladoga	3	13.30b	4.45	33	1.41- 4.23
Clinton	4	6.94b	4.45	64	1.41- 4.23
Grundy	2	1.11a	0.56	50	0.42- 1.41
Pershing	2	1.11a	0.56	50	0.42- 1.41
Weller	2	5.56a	2.50	45	0.42- 1.41
Mahaska var.§	2	31.10a	7.50	24	1.41- 4.23
Givin	2	5.00b	2.22	44	1.41- 4.23
Taintor	1	4.17a	0.83	20	1.41- 4.23
Rubio	1	2.78a	1.95	70	1.41- 4.23
Clarion	2	46.70a	13.60	29	4.23-14.10
Lester	2	16.90b	7.22	43	4.23-14.10
Hayden	3	15.60b	11.40	73	4.23-14.10
Nicollett	3	43.10a	19.20	45	4.23-14.10
LeSueur var.§	1	14.70b	6.95	47	4.23-14.10
Webster	2	21.10a	10.60	50	4.23-14.10
Cordova	1	25.30a	10.80	43	1.41- 4.23
Ames	2	5.00b	3.33	67	1.41- 4.23
Kenyon	3	30.30a	8.60	28	4.23-14.10
Bassett	1	21.90a	10.60	48	4.23-14.10

* Percolation rates within a biosequence followed by the same letter do not differ significantly at the 0.05 level as determined by an *F*-test (Snedecor and Cochran, 1967).

† Soil series are grouped by biosequence.

‡ Estimated hydraulic conductivity of 0.76- to 1-m depth from Soil Conservation Service SOI-5 Soil Interpretation Records.

§ Clay content in subsoil is slightly below the range established in the official series description.

Table 5. Morphological characteristics of soil biosequences.

Soil series	Clay content, %	Bulk density, Mg m^{-3}	Occurrence of argillans†	Structure‡	Moist consistence§	Horizon boundary¶
Tama	28.1a**	1.37a	1	1,m,pr/1,m,sbk	1	g
Downs	32.8ab	1.40b	1	2,m,sbk	1	c
Fayette	33.9b	1.43b	2	1,c,pr/1,m,abk	1	c
Otley	35.8a	1.32a	3	2,m,sbk	2	g
Ladoga	35.7a	1.40b	3	2,m,sbk	2	g
Clinton	37.2b	1.46b	3	2,m,pr/2,m,sbk	2	c
Grundy	36.3a	1.47a	4	2,m,pr/2,m,sbk	2	g
Pershing	36.1a	1.46a	2	2,m,sbk	2	g
Weller	38.4a	1.51a	3	1,m,sbk	3	g
Mahaska var.#	34.0a	1.43a	3	2,f+m,sbk	1	g
Givin	33.2a	1.47a	3	1,m,pr	2	g
Taintor	38.0–	1.46–	3	2,m,pr/2,f,sbk	2	g
Rubio	39.3–	1.45–	4	1,m,pr/2,m,sbk	2	g
Clarion	17.4a	1.55a	1	1,c+m,sbk	1	g
Lester	25.4b	1.44b	3	1,f+m,sbk	2	g
Hayden	31.5b	1.56a	4	1,f+m,sbk	2	c
Nicollett	20.5–	1.64–	1	1,f,sbk+m	1	g
LeSueur var.#	25.0–	1.61–	2	1,f,pr/1,m,sbk	2	g
Webster	30.2a	1.57a	1	1,m,sbk	1	g
Cordova	25.0–	1.61–	2	1,m,pr/1,sbk	1	g
Ames	29.4a	1.67b	4	1,m,pr/1,f,sbk	2	c
Kenyon	23.8–	1.76–	1	2,m,sbk	2	g
Bassett	26.8–	1.93–	2	1,f,pr/1,m,sbk	2	g

** Clay content and bulk density within a biosequence followed by the same letter do not differ significantly at the 0.10 level as determined by an *F*-test (Snedecor and Cochran, 1967).

† Occurrence of argillans: 1 = none; 2 = thin discontinuous; 3 = thin continuous; 4 = thick continuous.

‡ Soil structure described in the order: grade, size, and type. Soil structure grade: 1 = weak; 2 = moderate; 3 = strong. Soil structure size: f = fine; m = medium; c = coarse. Soil structure type: pr = prismatic; sbk = subangular blocky; abk = angular blocky; / = breaking to.

§ Moist consistence: 1 = friable; 2 = firm; 3 = very firm.

¶ Distinctness of horizon boundary; g = gradual; c = clear; d = diffuse.

Clay content in subsoil is slightly below the range established in the official series description.

soil group, has no significant difference in percolation rate among biosequence members and coefficients of variation are similar in the prairie end-member, transition member, and forest end-member. A lack of significant difference in percolation rates of this biosequence suggests that the strong degree of profile development (i.e., similar clay content and bulk density in subsoil, thickness of argillans, and subsoil consistence) in the Grundy, Pershing, and Weller soils precludes native vegetation effects on percolation rate found in other less-strongly developed biosequences.

Percolation rates were compared to estimated subsoil permeabilities obtained from SOI-5 Soils Interpretation Records (Soil Conservation Service, 1985). In general, percolation rates of transition members and forest end-members of biosequences, Alfisols, fall within the range of estimated subsoil hydraulic conductivities (Table 4). Percolation rates of prairie (Mollisol) end-members of biosequences, however commonly exceed estimated subsoil hydraulic conductivity by a factor of two or three. Recognition of more permeable subsoil in prairie end-members of the Otley and Webster biosequences is reflected in greater estimated hydraulic conductivity (Table 4). More permeable subsoil in prairie end-members of the Tama, Mahaska, Clarion, and Nicollet biosequences is not recognized in SOI-5 Soil Interpretation Records. Based on percolation rates, subsoil permeability of prairie end-members of the Tama, Clarion, Mahaska, and Nicollet biosequences is underestimated in the SOI-5 Soil Interpretation Records. The lack of significant difference in percolation rates among biosequence members of the Grundy, Taintor, and Kenyon biosequences is consistent with trends of estimated subsoil hydraulic conductivity.

Differences in percolation rate of Alfisols and Mollisols may be related to orientation of translocated clay when clay maximum and average clay content in the subsoil are similar. Within a biosequence, Alfisols typically have stronger expression of cutans or argillans, translocated clay on ped surfaces, in channels, and in pores than do Mollisols (Table 5). As the translocated clay swells during wetting, it reduces the effective pore diameter and reduces permeability.

Developmental Sequences

The Tama-Otley-Grundy, Downs-Ladoga-Pershing, and Fayette-Clinton-Weller soil sequences are respectively, prairie end-member, transition member and forest end-member sequences of increasing profile development (Cain and Riecken, 1958). Percolation rates of Tama and Otley soils, developmental stage II and stage III soils, respectively, are similar and are significantly higher than the percolation rate of the Grundy soil, a developmental stage IV soil (Table 4). Percolation rates of Downs and Ladoga, developmental stage II and III soils, respectively, are similar and are significantly higher than the percolation rate of the Pershing soil a developmental stage IV soil. Degree or stage of profile development may have less of an effect on percolation rates in the forest end-member stage of development sequence, Fayette-Clinton-Weller. While there is a trend of decreasing percolation rate

Table 6. Percolation test results and estimated hydraulic conductivity for selected topo-drainage sequences.

Soil series†	n	Percolation test results			Estimated hydraulic conductivity‡
		Mean	Standard deviation	Coefficient of variation	
		$\text{m s}^{-1} \times 10^{-4}$		%	$\text{m s}^{-1} \times 10^{-4}$
Galva	2	51.40a*	6.95	14	4.23-14.10
Primghar	2	47.20a	4.17	9	4.23-14.10
Marcus	1	62.20a	14.20	23	1.41- 4.23
Sharpsburg	2	18.30a	6.11	33	4.23-14.10
Macksburg	3	15.60ab	12.50	80	4.23-14.10
Winterset	3	4.17b	1.95	47	1.41- 4.23
Otley	2	26.10a	9.45	36	4.23-14.10
Mahaska var.§	2	31.10a	7.50	24	1.41- 4.23
Taintor	1	41.70b	8.33	20	1.41- 4.23
Ladoga	3	13.30a	4.45	33	1.41- 4.23
Givin	2	5.00b	2.22	44	1.41- 4.23
Rubio	1	2.78b	1.95	70	1.41- 4.23
Clarion	2	46.70a	13.60	29	4.23-14.10
Nicollet	3	43.10a	19.20	45	4.23-14.10
Webster	2	21.10a	10.60	50	4.23-14.10
Lester	2	16.90a	7.22	43	4.23-14.10
LeSueur var.§	1	14.70a	6.95	47	4.23-14.10
Cordova	1	25.30a	10.80	43	1.41- 4.23
Hayden	3	15.60a	11.40	73	4.23-14.10
Ames	2	5.00a	3.33	67	1.41- 4.23
Kenyon	3	30.30a	8.61	28	4.23-14.10
Floyd	1	55.00a	19.70	36	1.41- 4.23
Clyde	1	36.10a	7.78	22	1.41- 4.23

* Percolation rates within a topo-drainage sequence followed by the same letter do not differ significantly at the 0.05 level as determined by an *F*-test (Snedecor and Cochran, 1967).

† Soil series are grouped by topo-drainage sequence.

‡ Estimated hydraulic conductivity of 0.76- to 1-m depth from Soil Conservation Service SOI-5 Soil Interpretation Records.

§ Clay content in subsoil is slightly below the range established in the official series description.

with increasing profile development there is no significant difference in percolation rate for the Fayette-Clinton-Weller developmental sequence. Lack of significant differences in percolation test results of this soil sequence may be related to the variability of percolation test results. Additional percolation test data may be needed to better characterize permeability of the forest end-member sequence of soils with increasing profile development.

The Tama-Otley-Grundy and Downs-Ladoga-Pershing developmental sequences display trends of increasing coefficients of variation with increasing profile development. The increasing coefficient of variation associated with increasing degree of profile development may in part be related to the increased distinctness of subsoil horizon boundaries associated with increasing degree or stage of profile development. Distinctness of subsoil horizon boundaries is similar in the Fayette-Clinton-Weller developmental sequence, and may account for their similar coefficients of variation.

Percolation rates of the transition member stage of profile development sequence, Downs-Ladoga-Pershing, and the forest end-member sequence, Fayette-Clinton-Weller, are within the range of estimated subsoil hydraulic conductivity (Table 4). Percolation rates of the prairie end-members, Tama-Otley, stage of profile development sequence exceed estimated subsoil hydraulic conductivity by a factor of two or three while the Grundy soil has a percolation rate within the range of estimated hydraulic conductivity.

Table 7. Morphological characteristics of soil topo-drainage sequences.

Soil series	Clay content, %	Bulk density, Mg m ⁻³	Occurrence of argillans†	Structure‡	Moist consistence§	Horizon boundary¶
Galva	28.1a**	1.32a	1	1,f,pr/1,f,sbk	1	c
Primghar	24.6a	1.28a	1	1,f,pr	1	g
Marcus	26.9-	1.42-	1	1,f,pr	1	g
Sharpsburg	35.7a	1.32a	2	2,m,pr	1	d
Macksburg	35.6a	1.39ab	3	3,m,pr/3,m+f,sbk	1	g
Winterset	35.7a	1.48b	2	1,m,pr/2,m+f,sbk	2	g
Otley	35.8a	1.32a	3	2,m,sbk	2	g
Mahaska var.#	34.0a	1.43a	3	2,f+m,sbk	1	g
Taintor	38.0-	1.45-	4	1,m,pr/2,m,sbk	2	g
Ladoga	35.7a	1.40a	2	2,m,sbk	2	g
Givin	33.2a	1.47a	3	1,m,pr	2	g
Rubio	39.3-	1.45-	3	1,m,pr/2,m,sbk	2	g
Clarion	14.7a	1.55a	1	1,c+m,sbk	1	g
Nicollett	20.5a	1.64b	1	1,f,sbk	1	g
Webster	30.2b	1.57a	1	1,m,sbk	1	g
Lester	25.4-	1.44-	3	2,f+m,sbk	2	g
LeSueur var.#	25.0-	1.61-	2	1,m,pr/1,m,sbk	2	g
Cordova	25.0-	1.61-	2	1,m,pr/1,f,sbk	2	g
Hayden	31.5a	1.56a	4	1,f+m,sbk	2	c
Ames	29.4a	1.67b	4	1,m,pr/1,f,sbk	2	c
Kenyon	23.8-	1.76-	1	2,m,sbk	2	g
Floyd	29.2-	1.83-	1	1,m,pr/1,m,sbk	2	c
Clyde	28.5-	1.73-	1	1,m,sbk	2	g

** Clay content and bulk density within a topo-drainage sequence followed by the same letter do not differ significantly at the 0.10 level as determined by an *F*-test (Snedecor and Cochran, 1967).

† Occurrence of argillans: 1 = none; 2 = thin discontinuous; 3 = thin continuous; 4 = thick continuous.

‡ Soil structure described in the order: grade, size, and type. Soil structure grade: 1 = weak; 2 = moderate; 3 = strong. Soil structure size; f = fine; m = medium; c = coarse. Soil structure type: pr = prismatic; sbk = subangular blocky; abk = angular blocky; / = breaking to.

§ Moist consistence: 1 = friable; 2 = firm; 3 = very firm.

† Distinctness of horizon boundary; g = gradual; c = clear; d = diffuse.

Clay content in subsoil is slightly below the range established in the official series description.

Topo-Drainage Sequences

Percolation rates of topo-drainage sequences of soils without an argillic horizon, developmental stage I and II Mollisols, display no relationship to natural soil drainage class. There is no significant difference in percolation rates among soils of the Galva, Clarion, and Kenyon topo-drainage sequences (Table 6). Within topo-drainage sequences of soils without an argillic horizon, there is no trend evident for coefficient of variation and natural soil drainage class.

Subsoil hydraulic conductivity estimates of developmental stage I and II Mollisols indicate differences in subsoil permeability for two of the three topo-drainage sequences examined (Table 6). Hydraulic conductivity is estimated to be slower in the poorly drained Marcus and somewhat poorly drained Floyd soils than in other developmental stage I soils of their respective soil sequence. Percolation rates of the Marcus and Floyd soils were similar to percolation rates of their respective topo-drainage sequence members, however. Galva, Clarion, and Kenyon topo-drainage sequences are end-members of prairie biosequences and their percolation rates exceed estimated subsoil hydraulic conductivity by a factor of two or three.

In soils with an argillic horizon (i.e. stage II and III soils) percolation rates of three topo-drainage sequences vary according to natural soil drainage and two do not (Table 6). Percolation rates of well- and moderately well-drained soils are significantly higher than percolation rates of poorly drained soils of the Sharpsburg, Otley, and Ladoga topo-drainage sequences but are not significantly different in the Lester and Hayden topo-drainage sequences. There did not

appear to be any consistent trend between natural drainage class and coefficient of variation. The Otley and Hayden topo-drainage sequences exhibit decreasing coefficients of variation as drainage becomes more restrictive while the Sharpsburg and Ladoga topo-drainage sequences exhibit higher coefficients of variation as drainage becomes more restrictive.

Morphological differences do not appear to account for the lower percolation rates of some poorly drained soils (Table 7). Poorly drained soils have a shallower depth to the water table than better drained members of the sequence and the lower percolation rate of poorly drained soils may be an artifact of the percolation test procedure. Poorly drained soils had higher antecedent moisture contents than well-drained and moderately well-drained soils when the test holes were dug (Luce, 1973; Bicki, 1977; DeWitt, 1978). Smearing of the sidewalls could occur to a greater extent under higher antecedent moisture contents and could result in reduced permeability. Sidewalls were roughed in an attempt to alleviate the effects of smearing. Luce (1973) and Bicki (1977) found no correlation between percolation rate and antecedent moisture content while DeWitt (1978) found percolation rate to be negatively correlated with antecedent moisture content.

In general, percolation rates of topo-drainage sequence members with an argillic horizon fall within the range of estimated subsoil hydraulic conductivity. With the exception of the Lester topo-drainage sequence, slower percolation rates in poorly drained members of topo-drainage sequences with an argillic horizon are consistent with slower estimated hydraulic conductivities assigned in the SOI-5 Soil In-

terpretation Records (Soil Conservation Service, 1985) (Table 6). Percolation rates of the Sharpsburg, Ladoga, and Hayden topo-drainage sequences are generally within the range of estimated subsoil hydraulic conductivity. Percolation rates of the moderately well- and somewhat poorly drained members of the Otley topo-drainage sequence exceed estimated subsoil hydraulic conductivity by a factor of two or three (Table 6).

CONCLUSIONS

Percolation test results for an array of biosequences and stage of development sequences in Iowa reflected differences in subsoil morphology. Within a biosequence, Mollisols had higher percolation rates and lower coefficients of variation than Alfisols. The lower percolation rates of Alfisols were attributed to higher subsoil clay content, higher subsoil bulk density, thicker more continuous argillans, and a stronger subsoil consistence in Alfisols. Percolation test results within topo-drainage sequences without an argillic horizon were similar and reflected similar morphological characteristics of topo-sequence members.

Percolation test results of Alfisols examined in this study approximate Soil Conservation Service (1985) estimates of subsoil hydraulic conductivity but coefficients of variation are quite high. Percolation test results of Mollisols examined in this study commonly exceed estimated hydraulic conductivities by a factor of two or three and have relatively low coefficients of variation. On the basis of percolation test results, estimated subsoil hydraulic conductivity of developmental stage I, II and III Mollisols examined in this study is more permeable than current SCS estimates. More permeability data is needed to quantify and more accurately estimate subsoil hydraulic conductivity. Our data suggests that more consideration be given to biosequence effects and stage of profile development and less consideration be given to topo-drainage sequence effects when assigning estimated hydraulic conductivity ranges in Soil Interpretation Records. Furthermore, criteria used to establish estimated hydraulic conductivity should be applied consistently to biosequences, topo-drainage sequences, and stage of profile development sequences.

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